



P3Q Corona Kind: Module-LWE Threshold Verify on the Unified Precompile Slot

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Abstract

LP-218 introduces P3Q at EVM slot 0x012205 as the PQ-verifier family for rollup-batch authentication, specifying the unified kind-byte dispatch ABI and the first verifier kind 0x01 (Pulsar / FIPS-204 ML-DSA-65 threshold). This paper specifies *kind 0x02 Corona*: a Module-LWE threshold verifier built on the two-round Corona construction (eprint 2024/1113). The shared P3Q framework — the unified slot, the kind-byte dispatch ABI, and the P3QVerifier registration interface — is specified in LP-219; this paper gives Corona’s wire layout and its gas calibration (5 000 000 gas). Corona reduces to Module-LWE (Ringtail/Raccoon lineage). It shares the Module-LWE *hardness family* with Pulsar (LP-223) but is a distinct scheme at a distinct parameter set: the dual-lattice posture forces an adversary to break *both* constructions — strictly stronger than either alone, though *not* hardness-family-disjoint, since both reduce to Module-LWE. Cross-family independence comes only from Magnetar’s hash-collision assumption (LP-222). Corona is the Module-LWE parameter-diversity leg of the LP-217 PQ-strict posture. We also place Corona in the DEX rollup settlement pipeline: per-market fill batches roll up to the Z-Chain and are verified on a two-rung ladder — a kind 0x01 ML-DSA-65 attestation rung (measured ~ 4714 verifies/s) and a **starkfri** STARK/FRI validity rung (LP-221, measured ~ 986 /s on a toy AIR) — with Corona as the optional Module-LWE second leg for value-bearing markets and bundled $K \times V \times$ markets settlement. GPU dispatch is projected, not measured. Corona’s threshold library has landed (**corona v0.10.x**; the dedicated Corona precompile at 0x012206 is wired); the *P3Q-family* kind-0x02 dispatch entry at 0x012205 remains reserved and returns **abiFalse** via **ErrKindNotImplemented** until that dispatch is wired.

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1 Introduction

P3Q (LP-219) is the unified-slot post-quantum verifier framework at EVM precompile slot 0x012205: one slot, one ABI, and a leading `kind` byte that dispatches into a per-kind threshold-signature verifier core. This paper specifies *kind 0x02 Corona*, the Module-LWE member of the P3Q family.

For the slot allocation, the kind-byte dispatch ABI, the `P3QVerifier` Go interface that registers each kind, and the cross-family composition into LP-217 cert profile modes, see LP-219. The sibling kinds are specified in LP-223 (Pulsar / rank- k Module-LWE) and LP-222 (Magnetar / hash-based FIPS-205 SLH-DSA).

Corona reduces to the hardness of Module-LWE (Ringtail/Raccoon lineage, built on the two-round construction of eprint 2024/1113). It shares the Module-LWE *hardness family* with its sibling Pulsar (kind 0x01) — the two differ in parameter regime and codebase, not hardness family, so pairing them buys construction- and parameter-level diversity (an adversary must break both), not hardness-family-disjoint insurance; that comes only from the hash-based Magnetar kind (kind 0x03, SHA-3 collision). Corona is the Module-LWE parameter-diversity leg of the LP-217 PQ-strict posture, calibrated at 5 000 000 gas. Performance numbers are tagged **measured** or **projected**. Corona’s threshold library has landed (`corona v0.10.x`; the dedicated Corona precompile at 0x012206 is wired); the *P3Q-family* kind-0x02 dispatch entry at 0x012205 remains reserved and returns `abiFalse` via `ErrKindNotImplemented` until that dispatch is wired, and GPU dispatch is projected, not measured.

2 Corona Kind (kind = 0x02)

2.1 Slot and ABI

0x012205 with leading `kind=0x02`. Same physical precompile address as Pulsar (kind 0x01) and Magnetar (kind 0x03); dispatch resolves to the Corona `P3QVerifier` implementation.

```
// EVM precompile at slot 0x012205, kind = 0x02
bool ok = P3Q.verify(
    0x02,                                // Corona kind
    coronaSig,                          // serialized Corona threshold sig
    messageHash,                        // H(prev_root || new_root || batch_hash)
    groupPubKey                         // Corona group public key
);
```

2.2 Underlying scheme

Corona [3] is the Lux deployment of a two-round lattice threshold signature scheme building on the Ringtail line [1]. Hardness reduces to Module-LWE over the ring $R_q = \mathbb{Z}_q[X]/(X^{256} + 1)$ at the Ringtail parameter set (module matrix $A \in R_q^{8 \times 7}$, 48-bit prime q), via the standard two-round distributed signing protocol.

Dealerless key generation. Corona is *natively dealerless*. Key generation runs a Pedersen-style verifiable-secret-sharing DKG over R_q (the `keyera` bootstrap over the shared `luxfi/dkg` VSS: commitment $\mathbf{C} = \mathbf{A} \cdot \text{NTT}(c) + \mathbf{B} \cdot \text{NTT}(r)$, commitment broadcast, a digest cross-check round, and verification with signed-complaint identifiable abort), noise-flooded so the group key is a true $\mathbf{A}\mathbf{s} + \mathbf{e}''$; no single party ever holds the group secret. As of `corona v0.10.x` this is the production keygen default (`keyera.Bootstrap`), now consuming the shared `luxfi/mlwe` Module-LWE base. Signing likewise never reconstructs the key: each party emits a *masked* partial $\mathbf{z}_j$ from its own share and the combiner sums them ($\mathbf{z} = \sum_j \mathbf{z}_j$), so the master secret $\mathbf{s} = \sum_j \lambda_j \mathbf{s}_j$ appears only as the coefficient of the public challenge in the masked aggregate, never formed.

Pulsar is now also dealerless. Its production FIPS-204 keygen uses Mithril short replicated secret sharing [2] — the published escape from the additive- S_η wall that earlier forced a trusted dealer — and its signing is the no-reconstruct hyperball signer (LP-073 [4]). So in the Quasar dual-PQ AND-mode certificate (LP-020) *both* lattice legs are dealerless: Corona composes alongside Pulsar as **defence-in-depth**, not as the sole carrier of the permissionless-public guarantee. Corona’s dealerlessness is unconditional; Pulsar’s carries a labeled FIPS-204 KeyGen distribution-equivalence residual (verifier-compatibility is proven), which is one more reason a sovereign validator set runs both legs.

Corona’s signature is *not* byte-equal to any NIST standard. Both Corona and Pulsar (FIPS-204 ML-DSA-65) are Module-LWE schemes; they differ in *parameter regime and codebase*, not in hardness family. Corona uses the Ringtail-derived parameter set (~ 33 KB signatures, NIST Cat-1) with Module-SIS polynomial commitments; Pulsar uses the ML-DSA-65 tuning (~ 3.3 KB, Cat-3, byte-equal to FIPS-204). Pairing them gives parameter- and implementation-level diversity — a break specific to one parameter set or codebase need not hit the other — but *not* family-disjoint hardness: a structural break of Module-LWE would threaten both. Cross-family independence on the signature lane comes only from the hash-based Magnetar kind (kind 0x03, SLH-DSA).

2.3 Wire format

Offset	Size	Field
0	1	kind = 0x02
1	1	mode = 0x01 (reserved Corona parameter set)
2	4	sigLen (BE32)
6	sigLen	coronaSig ($\sim 33\,052$ B)
6+sigLen	4	pkLen (BE32)
10+sigLen	pkLen	groupPubKey
10+sigLen+pkLen	32	messageHash

Table 1: Corona-kind wire format. Sig size is the dominant cost (~ 33 KB); the rest mirrors the Pulsar layout to keep parsing uniform across kinds.

The mode byte is reserved at 0x01; future Corona parameter set additions (a higher-security Module-LWE variant) get new mode bytes without changing the wire-format skeleton.

2.4 Verifier core

The Corona verifier core IS `luxfi/corona/sign.Verify`, and it is wired today behind the *dedicated* Corona precompile at slot 0x012206 (`precompile/corona/contract.go`, which imports `luxfi/corona/sign` and `luxfi/corona/threshold`). The core is bespoke — not FIPS-204 byte-equal — because Corona is a separate Module-LWE construction (Ringtail/Raccoon lineage), not a NIST standard.

Status: the Corona threshold library has landed (`corona v0.10.x`) and the dedicated Corona precompile at 0x012206 is wired. The *P3Q-family* kind-0x02 dispatch at the unified slot 0x012205 is what remains pending: the reference `p3q.Run` still routes `kind=0x02` to `abiFalse` via the `ErrKindNotImplemented` path,

```
case KindCorona, KindMagnetar:
    // Reserved kinds -- P3Q dispatch not yet wired. Return abiFalse
    // ...
    return abiFalse(), remaining, nil
```

so the verifier is callable today at 0x012206 but not yet via the unified P3Q slot. Solidity callers MAY emit `P3Q.verify(0x02, ...)` from network genesis; the call charges the per-kind

gas (see below) and returns `false` until the kind-0x02 dispatch is wired. This is the LP-218 “no revert on cryptographic failure” contract: `false` on the return word is the universal signal for “not verified for any reason,” and reserved-but-not-wired joins “malformed input” and “FIPS reject” under that umbrella.

2.5 Gas cost

5 000 000 gas – $\sim 100\times$ the Pulsar kind’s 50 000. Returned by the Corona `P3QVerifier.GasCost()` entry. Why the disparity?

1. **CPU verify cost is heavier.** The Corona verifier requires NTT inverse over the Module-LWE ring + polynomial multiplication mod q over the larger Ringtail parameter set (~ 33 KB sig vs Pulsar’s 3.3 KB).
2. **Calibration uses the CPU verify cost as the baseline.** LP-218’s calibration discipline assumes a CPU-class verifier – validators are not required to have Blackwell GPUs. Under that assumption, Corona’s effective per-validator verify cost is the parity audit’s measured ~ 1.6 ms CPU, which is approximately $20\times$ the Pulsar CPU verify time ($\sim 80\mu\text{s}$). The gas charge is rounded up to $100\times$ Pulsar’s for conservative accounting: 5 M gas.
3. **Block-budget headroom.** At 12 M block gas and 5 M per Corona verify, the limit is ~ 2 Corona verifies per block – this is the bottleneck for very high- density PQ-strict rollup throughput. The cost cliff is intentional; the LP-217 mode ladder uses it to price cross-family insurance correctly.

Operators with all-Blackwell validator sets may petition for a **Blackwell-discounted Corona gas charge** in a future amendment; LP-220 does not codify it. The conservative CPU-baseline gas charge applies until validator-hardware uniformity is established. This is the same calibration discipline LP-218 uses for the Pulsar kind.

2.6 Performance

Metric	M1 Max	Spark	Source
Per-validator Corona sign at $N=64$	~ 30 ms	~ 30 ms	matrix bench (measured) [7]
CPU verify (single sig)	~ 1.6 ms	~ 1.6 ms	parity audit (measured)
Blackwell sm_120 verify	—	~ 15 ms	projected (LP-203 kernel) [5]
Blackwell sm_120 batched ($N=64$)	—	~ 50 ms aggregate	projected

Table 2: Corona kind performance. Sign and CPU-verify times are measured per the parity audit; Blackwell verify timings remain projections until the GPU dispatch boundary closes (see LP-223).

2.6.1 Scaling exponent

The parity audit measured Corona-leg signing cost growing super-linearly in N . The measured exponent across $N \in \{16, 32, 64\}$ is approximately $T(N) \approx c \cdot N^{1.5-1.7}$, depending on the host platform. This is not an $O(N)$ scheme and should not be described as such in downstream documents; the Module-LWE distributed signing protocol’s full-rank check and Gaussian-hash PRF carry costs that grow faster than linearly in the participant count. The measured exponent is the operative number for sequencer capacity planning, not the naive-amortization figure.

2.7 Use case

A rollup that wants **intra-lattice parameter and implementation diversity** configures its sequencer to sign each batch with both a Pulsar threshold sig AND a Corona threshold sig —

two Module-LWE schemes at distinct parameter sets and codebases. The rollup contract on Z-Chain calls `P3Q.verify(0x01, ...)` and `P3Q.verify(0x02, ...)` at the same precompile address `0x012205` in sequence inside its `validateBatch` function; both must return `true` for the batch to be accepted.

A break specific to Pulsar’s parameter set or codebase need not compromise Corona’s, so the rollup retains a valid PQ leg under an implementation- or parameter-localized failure. This is *not* family-disjoint hardness: a structural break of Module-LWE itself would threaten both legs. For cross-family insurance the rollup must add the hash-based Magnetar kind (`P3Q.verify(0x03, ...)`). The cost is $\sim 5\,050\,000$ gas per batch (Pulsar $50\,000$ + Corona $5\,000\,000$). Acceptable for value-bearing rollups (custody, treasury, regulated securities); prohibitive for high-volume rollups (gaming, social).

This is the LP-217 PQ-strict posture made operationally callable.

3 Application: DEX Rollup Settlement to Z-Chain

The DEX matching path (Lux Lightspeed DEX [8]) fills orders at hundreds of millions per second per GPU on the D-Chain, far above any single chain’s finality rate. To anchor that throughput trustlessly, per-market fill batches roll up to the Z-Chain and are verified there through the P3Q family at slot `0x012205`. This section places Corona (kind `0x02`) in that settlement pipeline; the measured verify rates belong to the sibling kinds, and Corona is the optional Module-LWE parameter-diversity leg.

3.1 Two verification rungs

DEX rollup settlement uses a two-rung ladder, both callable at the same P3Q slot:

1. **Attestation rung (venue-trust).** The sequencer signs each per-market batch with a P3Q kind `0x01` (Pulsar / FIPS-204 ML-DSA-65) threshold signature. The Z-Chain contract calls `P3Q.verify(0x01, ...)` to confirm the batch was produced by the venue’s threshold quorum. This rung trusts the venue’s signer set but is cheap: the ML-DSA-65 verify path is measured at ~ 4714 verifies per second.
2. **Validity rung (trustless).** For trust-minimized settlement the sequencer additionally produces a STARK/FRI validity proof of correct matching, verified by the `starkfri` verifier (LP-221 [6]) — a cSHAKE256 / Goldilocks PQ Plonky3-fork verifier. This rung proves the fills are correct without trusting the signer set. The current verifier is measured at ~ 986 proofs per second against a **toy AIR**; this figure is the operative number and is explicitly not a production-circuit measurement.

The attestation rung is fast and venue-trusting; the validity rung is slower and trustless. A venue chooses the rung (or runs both) per its trust requirements.

3.2 Where Corona fits

Corona (kind `0x02`) is the *Module-LWE cross-family insurance leg* for the attestation rung, not a separate settlement mechanism. A value-bearing DEX rollup that wants independence from the Module-LWE assumption underlying ML-DSA-65 signs each batch with *both* a Pulsar (kind `0x01`) and a Corona (kind `0x02`) threshold signature and requires both `P3Q.verify` calls to return `true`, exactly as in Section 2’s use case. A Module-LWE break that compromises the Pulsar attestation does not compromise the Corona attestation; the rollup’s settlement remains cryptographically anchored.

The cost discipline of Section 2 applies unchanged: Corona’s $5\,000\,000$ gas makes the second leg appropriate for value-bearing markets (custody, treasury, regulated-securities venues) and prohibitive for high-volume markets, which settle on the kind `0x01` attestation rung alone (optionally backed by the `starkfri` validity rung). As noted throughout this paper, Corona’s GPU

dispatch is projected and its *P3Q* kind-0x02 dispatch is not yet wired (the threshold library has landed and the dedicated Corona precompile at 0x012206 is live). A DEX settlement path that wants a second Module-LWE scheme (distinct parameter set and codebase, not a distinct hardness family) uses the measured kind 0x01 rung today, with Corona reachable at 0x012206 and reserved at the *P3Q* kind-0x02 entry until that dispatch is wired.

3.3 Bundled settlement: $K \times V \times \text{markets}$

Per-fill, per-venue verification does not scale: at the attestation-rung's ~ 4714 verifies per second, a chain that verified one signature per fill would cap settlement far below the matcher's fill rate. The settlement model therefore bundles. A single Z-Chain settlement object aggregates K signer attestations across V venues across many markets into one verification event:

- **K signers.** The threshold quorum per venue contributes K partial signatures aggregated into one threshold signature that verifies in a single `P3Q.verify` call.
- **V venues.** Independent venues' batches are aggregated into one settlement object rather than verified one venue at a time.
- **Many markets.** Each venue's per-market arenas (Lux Lightspeed DEX) contribute their batch roots to the bundle, so a single verification covers many books.

Because one verification amortizes over $K \times V \times \text{markets}$ fills, the effective settled-fill rate at fleet scale is **projected** to exceed 10^9 fills per second of finality — this is a fleet-aggregation projection, not a single-verifier measurement. The per-verifier measured rates ($\sim 4714/s$ attestation, $\sim 986/s$ validity on a toy AIR) are the operative numbers; the $>10^9$ figure is what those rates compose to once the bundle factor $K \times V \times \text{markets}$ is applied across a multi-node Z-Chain settlement fleet. No single *P3Q* verifier reaches that rate.

References

- [1] Cecilia Boschini, Darya Kaviani, Russell W. F. Lai, Giulio Malavolta, Akira Takahashi, and Mehdi Tibouchi. Ringtail: Practical two-round threshold signatures from LWE. Cryptology ePrint Archive, Paper 2024/1113; IEEE Symposium on Security and Privacy (S&P) 2025, 2024. Academic two-round lattice threshold (IACR ePrint 2024/1113, IEEE S&P 2025). Not a Lux artifact; do not relabel as "Corona" or "Pulsar".
- [2] Andrea Celi, Rafaël del Pino, Thomas Espitau, Guilhem Niot, and Thomas Prest. Threshold ML-DSA from short replicated secret sharing (Mithril). In *USENIX Security*, 2026. <https://eprint.iacr.org/2026/013>. Short replicated secret sharing: the published escape from the naive-additive- S_η wall, making dealerless threshold ML-DSA keygen reachable under the stock FIPS-204 verifier.
- [3] Lux Industries. Corona: Two-round module-lwe threshold signatures. <https://github.com/luxfi/corona>, 2026. Lux's own threshold scheme: Module-LWE (Ringtail/Raccoon lineage), a rank- 8×7 module over $R_q = \mathbb{Z}_q[X]/(X^{256} + 1)$, $q = 0x1000000004A01$ — not Ring-LWE (rank 1). Two-round, natively dealerless (Pedersen-VSS, noise-flooded), no-reconstruct signing; consumes the shared `luxfi/mlwe` base. Builds on the Ringtail line ([1]); cited via this Lux artifact, never via the academic paper. Companion hardness is Module-SIS.
- [4] Lux Industries. Lp-073: Pulsar two-round lattice threshold signing, 2026.
- [5] Lux Industries. Lp-203: Gpu-native verify, 2026.
- [6] Lux Industries. Lp-221: starkfri — cshake256 / goldilocks pq plonky3-fork stark/fri verifier, 2026.

- [7] Lux Industries. Quasar 4-Scheme Matrix Bench – m1 max + dgx spark gb10, 2026. `lux/threshold/protocols/parity/parity_test.go`, run 2026-06-04. Mean of $N=\{16, 32, 64\}$ across 50 rounds.
- [8] Lux Industries, Inc. Lux lightspeed dex: On-chain order book with sub-second finality. Lux Technical Report.